Worksheet: Adiabatic gas expansion between two tanks

Name(s) __________________________________

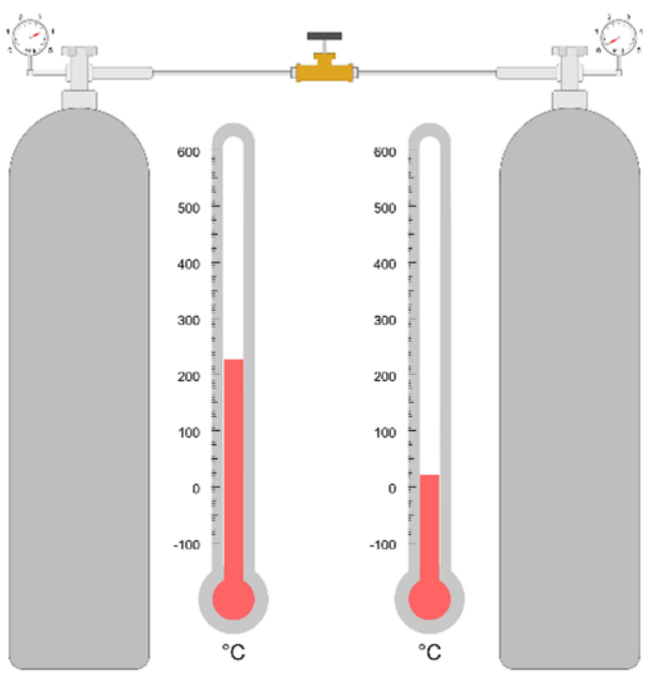
In this experiment, an ideal gas at high pressure in a tank expands adiabatically into a second tank that is under vacuum by opening a valve between the two tanks. The valve is closed as soon as the pressure equalizes between the two tanks. An energy balance is applied to the entire system, which is closed.

Student learning objectives

1. Apply the ideal gas law.
2. Do energy balances on a closed system.
3. Do mass balances.

Experimental Diagram

An insulated line that contains a valve connects the two tanks. Each tank has a valve, a pressure gauge, and a temperature sensor.



Assumptions

- The tanks and the line connecting them are well insulated.
- The volumes of the valves and the line connecting the tanks are insignificant.
- Gas expansion is fast so heat transfer from the gas to the tanks is insignificant, and thus the heat capacity of the tank material can be ignored.
- The gas is ideal, and it has a constant heat capacity C_P of 30. J/ (mol K) and $C_V = C_P$ - R.

Notation

n_{1i} = initial number of moles in first tank

C_V = constant volume heat capacity of gas

T_{1i} = initial temperature in tank 1

V_1 = tank 1 volume

P_{1i} = initial pressure in tank 1

T_{ref} = reference temperature

T_{1f} = final temperature in tank 1

n_{1f} = final number of moles in tank 1

P_{1f} = final pressure in tank 1

V_2 = tank 2 volume

n_{2f} = final number of moles in tank 2

T_{2f} = final temperature in tank 2

P_{2f}= final pressure in tank 2

$T_{average}$ = average final temperature

R = ideal gas constant [either units of (bar L)/(mol K)) or J/(mol K)]

C_P = constant pressure heat capacity

Questions to answer before beginning the experiment

1. If the two tanks are the same volume and the initial temperature in the first tank is 400 K, is the final temperature in the first tank greater than, less than, or equal to 400 K? Explain.

2. If the two tanks are the same volume and the initial temperature in the first tank is 400 K, is the final temperature in the second tank greater than, less than, or equal to 400 K? Explain.

3. If the two tanks are the same volume and the initial pressure in the first tank is 20 bar, is the final pressure in the system greater than, less than, or equal to 10 bar? Explain.

4. If the two tanks have different volumes and the initial pressure in the first tank is 20 bar, and the initial temperature is 400 K, is the final average temperature of the two tanks greater than, less than, or equal to 400 K? Explain.

5. If the two tanks are the same volume, is the final number of moles larger in the first or second tank. Why?

Run the experiment

1. Select the initial pressure, temperature, and volume of the first tank and record them in the table below.
2. Use the ideal gas law to calculate the initial number of moles in the first tank and record it in the table.

3. Select the volume of the second tank, which is under vacuum, and record it in the table.

Quantity	Tank 1 initial	Tank 1 final	Tank 2 initial	Tank 2 final
Pressure (bar)	P_{1i}=	P_{1f}=	0	P_{2f}=
Temperature (°C)	T_{1i}=	T_{1f}=	----	T_{2f}=
Volume (L)	V_1=	V_1=	V_2=	V_2=
Moles	n_{1i}=	n_{1f}=	0	n_{2f}=

4. Open the valve on the top of each tank by clicking on each valve. Then open the valve connecting the tanks by clicking on it. When the pressure in the second tank stops increasing and equals the pressure in the first tank, click the connecting valve to close it. Record the final pressure in the table.
5. Record the final temperatures in the tanks in the table.
6. Use the ideal gas law to determine the final number of moles in each tank and record them in the table.

7. Calculate the total number of moles in the two tanks. How close is this value to the initial number of moles in the first tank?

   $n_{1f} + n_{2f} =$ ____________ Percent difference ____________

8. Calculate the average final temperature $T_{average}$ of the system from an internal energy balance.

   $$(n_{1f} + n_{2f})C_V\left(T_{average} - T_{ref}\right) = n_{1f}C_V\left(T_{1f} - T_{ref}\right) + n_{2f}C_V\left(T_{2f} - T_{ref}\right)$$

   and a mass balance $\left(n_{1f} + n_{2f}\right) = n_{i1}$

   $T_{average}$ = ____________________

   Note that $T_{average}$ should equal T_{1i} because an internal energy balance on the system can also be written as:

   $$n_{1i}C_V\left(T_{1i} - T_{ref}\right) = n_{1f}C_V\left(T_{1f} - T_{ref}\right) + n_{2f}C_V\left(T_{2f} - T_{ref}\right)$$

   How close is the final average temperature to the initial temperature in tank 1?

   Percent difference ____________ Why?

9. How close is the final pressure to the pressure expected if the expansion were conducted isothermally?

   Percent difference ____________ Why?

10. Calculate the final temperature expected in the first tank by applying the equation below to a fraction of the moles initially in the tank (namely the fraction of gas that remains after expansion). This equation applies if the fraction of gas expanded adiabatically and reversibly:

    $$\frac{T_{1f}}{T_{1i}} = \left(\frac{P_{1f}}{P_{1i}}\right)^{R/C_P}$$

T_{1f} for adiabatic reversible expansion _______________

How close is the final temperature in the first tank to that expected for reversible adiabatic expansion?

Percent difference _____________ Why?

Questions to answer

1. Why did the temperature in the first tank increase/decrease?

2. Why did the temperature in the second tank increase above the initial temperature in the first tank?

3. How is the behavior of the system different if the initial pressure in the first tank was higher?

4. Did the specific entropy of gas in the first tank increase, decrease, or stay the same? Why?

5. Did the entropy of the system increase or decrease? Explain

6. What are sources of error in the measurements?

7. If this experiment were conducted in the laboratory, what might cause the measured pressure and temperatures to be different from the values calculated above?

8. What would change if the tanks were not insulated?

9. What safety measures would you employ if making this measurement in the laboratory?